WORKED SOLUTION

STEP II 1988, Question 11

Mechanics • String Winding Round Post

STEP II 1988, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s2-q11

The key idea

The free part of the string always leaves the post along a *tangent*. So the particle's position is "contact point plus a tangent vector of length (unwound string)", and that is easy to write down in terms of the angle θ. Differentiating, two terms cancel and the velocity turns out to be perpendicular to the string. The tension therefore does no work, the speed stays at its initial value, and the speed formula becomes a separable differential equation for θ.

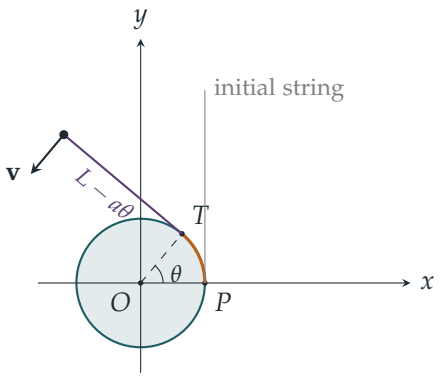
Solution

Setting up

We model the particle as a point mass on a smooth table, with a light inextensible string that stays taut. Gravity and the normal reaction are vertical and cancel, so the only horizontal force is the tension.

Take axes with origin O and put P at $(a, 0)$. The string starts perpendicular to OP, so it is tangent to the post at P, and we take the particle to start at (a, L). (The mirror-image choice just flips the sign of the y-coordinate.) As the string winds on anticlockwise, after it has wrapped through angle θ the length $a\theta$ lies along the arc from P, and the string leaves the post at

$$T = a(\cos\theta,\ \sin\theta).$$



The free part of the string, of length $L - a\theta$, runs along the tangent at T. The unit tangent pointing the way the string unwinds is $(-\sin\theta,\ \cos\theta)$: it is perpendicular to OT, and at $\theta = 0$ it is $(0, 1)$, matching the starting position.

Position and velocity

Hence the position vector of the particle is

$$\mathbf{r} = a\begin{pmatrix}\cos\theta\\ \sin\theta\end{pmatrix} + (L - a\theta)\begin{pmatrix}-\sin\theta\\ \cos\theta\end{pmatrix} = \begin{pmatrix}a\cos\theta - (L - a\theta)\sin\theta\\ a\sin\theta + (L - a\theta)\cos\theta\end{pmatrix}.$$

Differentiating with respect to t by the chain rule and the product rule,

$$\begin{aligned}\dot{\mathbf{r}} &= a\dot{\theta}\begin{pmatrix}-\sin\theta\\ \cos\theta\end{pmatrix} - a\dot{\theta}\begin{pmatrix}-\sin\theta\\ \cos\theta\end{pmatrix} + (L - a\theta)\dot{\theta}\begin{pmatrix}-\cos\theta\\ -\sin\theta\end{pmatrix}\\ &= -(L - a\theta)\dot{\theta}\begin{pmatrix}\cos\theta\\ \sin\theta\end{pmatrix}.\end{aligned}$$

The first two terms cancel: moving the contact point along the post is exactly undone by the shortening of the free string.

Since $(\cos\theta, \sin\theta)$ is a unit vector, $L - a\theta > 0$, and $\dot{\theta} > 0$ while the string winds on, the speed is

$$|\dot{\mathbf{r}}| = (L - a\theta)\dot{\theta},$$

as required.

Time to hit the post

Ask yourself The string is pulling on the particle, so why should its speed stay equal to v?

The velocity is parallel to $(\cos\theta, \sin\theta)$, while the string, and hence the tension, lies along $(-\sin\theta, \cos\theta)$. These are perpendicular, so the tension does no work. It is the only horizontal force, as the table is smooth. So the kinetic energy, and with it the speed, stays constant at v. The tension only changes the direction of the velocity.

Therefore, for all t until the particle reaches the post,

$$(L - a\theta)\,\frac{\mathrm{d}\theta}{\mathrm{d}t} = v, \qquad \theta(0) = 0.$$

Separating variables and integrating from 0 to t,

$$L\theta - \tfrac{1}{2}a\theta^2 = vt.$$

The particle hits the post when the whole string is wound on, that is, when $a\theta = L$, so $\theta = L/a$. Then

$$vt = \frac{L^2}{a} - \frac{L^2}{2a} = \frac{L^2}{2a}, \qquad \text{so} \qquad t = \frac{L^2}{2av},$$

which is the required time.

Checks

- At $\theta = 0$ the formula gives $\mathbf{r} = (a, L)$ and $\dot{\mathbf{r}} = (-L\dot{\theta}, 0)$, perpendicular to the initial string, as it should be.
- Units: $L^2/(av)$ is $\mathrm{m}^2/(\mathrm{m}\cdot\mathrm{m\,s^{-1}}) = \mathrm{s}$.
- Another way to see the answer: the particle's whole path has length

  $$\int_0^{L/a} (L - a\theta)\,\mathrm{d}\theta = \frac{L^2}{2a},$$

  travelled at constant speed v.
- We simulated the motion numerically, with a very stiff string wrapping round a post with $a = 1$, $L = 3$, $v = 2$. The simulated time matched $L\theta - \frac{1}{2}a\theta^2 = vt$ to within 10^{-5}, and integrating $\dot{\theta} = v/(L - a\theta)$ to $a\theta = L$ gave 2.25, which is $L^2/(2av)$.

Common slips

- Using conservation of angular momentum about O. The tension does not pass through O (it acts along a tangent), so angular momentum about O is *not* conserved. Energy is the right tool.
- Writing the free string along $(\cos\theta, \sin\theta)$, i.e. radially, instead of along the tangent.
- Forgetting that the speed formula needs $\dot{\theta} > 0$. The absolute value of the vector is $(L - a\theta)|\dot{\theta}|$.

Beyond STEP

The path traced out by the particle is an *involute* of the circle, the curve traced by the end of a taut thread unwound from a spool. Involutes are why gear teeth are shaped as they are, and they were central to Huygens' work on pendulum clocks. Its arc length

$\int (L - a\theta)\,\mathrm{d}\theta$ is a first example of the differential geometry of plane curves (evolutes, involutes and curvature). For the mechanics side, including when energy and angular momentum are or are not conserved, read Kleppner and Kolenkow, *An Introduction to Mechanics*.

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